

Machine Learning Engineer specializing in speech recognition, applied NLP, and cloud deployment pipelines. Proven record of managing the full ML lifecycle, from dataset curation and GPU training to building highly efficient real-time production systems. Developed and deployed True Tilawah, an Arabic speech AI system achieving 90%+ domain accuracy, optimizing complex multistage audio inference streams down to sub-3-second latencies. Expert in optimizing open-source models, constructing stateless WebSockets.

90%+ ASR Accuracy

Achieved on domain-specific Arabic transcription, beating standard baselines by 20%.

< 3s Global Latency

Optimized 6-stage audio streaming pipeline via stateless FastAPI WebSockets.

100% Client Sign-Off

Shipped production-grade ERP core modules.

TECHNICAL SKILLS

DEEP LEARNING & SPEECH AI

Speech Recognition (ASR), Model Fine-Tuning, Transfer Learning, OpenAI Whisper, HuggingFace Transformers, PyTorch, Scikit-Learn, WER Evaluation, GPU Training (T4).

NLP & AUDIO PROCESSING

Arabic ASR, Real-Time Signal Processing, Voice Activity Detection (Silero VAD ONNX), Fuzzy String Matching (RapidFuzz), Inverted Index Search, NLTK.

DEPLOYMENT & SYSTEMS ARCHITECTURE

Real-Time AI Pipelines, WebSockets, FastAPI, REST APIs, Stateless Architectures, Railway Cloud Deployment, Docker, Workflow Automation, ffmpeg.

LANGUAGES, DATABASES & TOOLS

Python, JavaScript, SQL, VB.NET, Git, GitHub, MySQL, Linux Shell, Google Colab Infrastructure.

PROFESSIONAL EXPERIENCE

AI/ML Engineer (Final Year Project) — TRUE TILAWAH

2024 – Present | Lahore, Pakistan

PYTORCH

OPENAI WHISPER

FASTAPI

WEBSOCKETS

SILERO VAD

RAILWAY CLOUD

- End-to-End AI Engineering:** Designed, fine-tuned, and shipped a live Arabic speech-recognition platform, improving Word Error Rate (WER) to deliver a 20 percentage-point accuracy gain over OpenAI's baseline.
- Production AI Pipeline:** Engineered an asynchronous 6-stage pipeline (VAD → ASR → Verse Matching → Phonetic Word Scoring → Rules Assessment → JSON Output Generator) reducing full-cycle feedback latency to <3 seconds.
- GPU Infrastructure & Fine-Tuning:** Orchestrated model transfer learning on an indexed custom Quranic audio dataset of 6,236 distinct verses utilizing hardware-accelerated T4 compute configurations.
- Algorithmic Optimizations:** Embedded Silero VAD (ONNX) for localized speech frame segmentation and built a sub-second text matching mechanism using a custom RapidFuzz inverted index matrix.
- Scalable Cloud Deployment:** Built and containerized a fully stateless backend framework deployed onto Railway, enabling horizontal autoscaling with minimal per-user infrastructure cost.

ERP Developer Intern — NOUVELLE SOFTWARE HOUSE

06/2024 – 09/2024 | Pakistan

VB.NET

MYSQL

UNIT TESTING

- Enterprise Software Delivery:** Architected and delivered core transactional modules for a multi-tenant ERP suite within a strict 3-month window, automating high-throughput inventory control and financial workflow loops.
- Database Integration & Optimization:** Authored optimized relational database queries using VB.NET and MySQL to support instantaneous data state changes across distributed operations modules.
- Production Quality:** Adhered to strict code compliance and systematic unit testing, securing full enterprise sign-off on initial submission with 0 subsequent bug tickets or engineering rework.

FEATURED PROJECTS

True Tilawah: Production Speech AI & Feedback Platform

[GitHub](#) | [Demo Video](#)

HUGGINGFACE

/RAPIDFUZZ

FFMPEG

GROQ API

DOCKER

Architected a full-scale AI application providing real-time acoustic feedback. Developed a custom transfer-learned Whisper speech transformer, hooked directly to multi-threaded WebSockets to perform simultaneous audio stream chunking, alignment mapping, and linguistic variance checking.

EDUCATION

Bachelor of Computer Science (BSCS)

Graduated Complete

University of Central Punjab (UCP), Lahore, Pakistan

Focus Electives: Machine Learning, Deep Learning, Natural Language Processing (NLP), Advanced Algorithms, Relational Database Systems.